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import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind

# Control group
control = np.array([70, 72, 68, 75, 71, 69, 73, 70, 74, 72])

# Treatment group
treatment = np.array([80, 82, 79, 85, 83, 81, 84, 80, 86, 82])

# Perform independent t-test
t_stat, p_value = ttest_ind(control, treatment)

print("Control Group Mean:", round(control.mean(), 2))
print("Treatment Group Mean:", round(treatment.mean(), 2))
print("T-statistic:", round(t_stat, 4))
print("P-value:", round(p_value, 6))

if p_value < 0.05:
    print("Reject Null Hypothesis")
    print("Treatment has a statistically significant effect.")
else:
    print("Fail to Reject Null Hypothesis")
    print("Treatment does not have a statistically significant effect.")

# Visualization
groups = ["Control", "Treatment"]
means = [control.mean(), treatment.mean()]

plt.bar(groups, means)
plt.ylabel("Average Outcome")
plt.title("Control vs Treatment")
plt.show()
```